

Machine Learning for Natural Language Understanding

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Transformers

Lecture 6 – Encoder

Sources Lecture

- Transformers
 - <https://youtu.be/ptuGIIU5SQQ?t=128> (Intro)
 - <https://youtu.be/ptuGIIU5SQQ?t=275> (Issues with RNNs)
 - <https://youtu.be/ptuGIIU5SQQ?t=826> (intra-sentence attention)
 - <https://youtu.be/ptuGIIU5SQQ?t=1442> (barriers and solutions of self-attention)
 - <https://youtu.be/ptuGIIU5SQQ?t=2191> (Transformer model)
 - <https://youtu.be/ptuGIIU5SQQ?t=2781> (Tricks for Training)
 - <https://youtu.be/ptuGIIU5SQQ?t=3489> (Decoder cross-attention)
 - <https://youtu.be/ptuGIIU5SQQ?t=3902> (empirical results)
- <https://web.stanford.edu/class/cs224n/>

Flipped Classroom Questions

- Which context does a W2V embedding capture compared to an RNN embedding?
- What is intra-sentence attention?
- What context does a Bert embedding capture?
- What does an attention weight in Bert do?

“The future of language is about the development of language models...”¹

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